A Schur/Frobenius Hybrid Envelope

A Sign-Free Finite Bound for Masked Twin-Prime Responses

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September 2, 2026

Abstract

The support-restricted Frobenius envelope in TPC-339 is universally valid but loose on broad arithmetic masks. We combine it with a global Schur envelope for the same finite symmetric all-plus shell operator. If x is supported on S and R is the maximum absolute row sum, then

$$\|Ax\|_2^2 \le \min\{\|A_{[:,S]}\|_F^2, R^2\}\|x\|_2^2.$$

The inequality is proved exactly for every finite symmetric matrix and passes all 216 records in the parent-locked panel. The Schur branch is active in 54 records and improves the zero-support envelope by factors from 1.250245 to 4.698443. Broad-mask occupancy nevertheless remains below 0.186855, so the hybrid is a useful finite envelope and a sharper obstruction, not a growing arithmetic estimate.

1 Question and finite object

TPC-338 showed that signed class covariances depend on the chosen control ensemble, and TPC-339 replaced that unstable interface by a support-only Frobenius inequality. We ask whether a second sign-free norm can sharpen the finite envelope without importing a covariance sign.

The parent-locked source is the four-way decomposition

$$\beta = \beta_T + \beta_B + \beta_P + \beta_Z,$$

where the labels denote twin-prime, non-twin-prime-shift, prime-power-shift, and zero-support masks. We use the all-plus deleted-diagonal shell operator

$$A = \sum_{54<p\le 108} B_{p,54,1}, \qquad Q = 54, \quad H = 66,$$

on

$$I_{o,N} = \{o, \dots, o + N/2 - 1\}, qquado \in \{42001, 44001\}, \quad N \in \{2048, 4096, 8192\}.$$

The nine controls are the identity, reversal, and the seven affine maps declared in TPC-338. A control only permutes source coordinates; the matrix and source construction are otherwise unchanged.

2 Two sign-free bounds

Let $S = \mathrm{supp}(x)$ and let A_S be the column submatrix. The support-restricted bound used by TPC-339 is

$$\|Ax\|_2^2 = \|A_S x_S\|_2^2 \le \|A_S\|_F^2 \|x_S\|_2^2. \tag{1}$$

Write

$$F(S)^2 = \|A_S\|_F^2 = \sum_{t\in S}\sum_u |A(u,t)|^2.$$

For a finite symmetric matrix put

$$R = \max_i \sum_j |A(i,j)|.$$

Symmetry gives $\|A\|_1 = \|A\|_\infty = R$, and the induced-norm inequality therefore gives

$$\|Ax\|_2^2 \le \|A\|_2^2 \|x\|_2^2 \le R^2 \|x\|_2^2. \tag{2}$$

Taking the smaller right-hand side in (1) and (2) proves the hybrid envelope

$$\|Ax\|_2^2 \le \min\{F(S)^2, R^2\} \|x\|_2^2. \tag{3}$$

No sign of a matrix entry or a cross-class covariance appears in this argument.

Exact anchor. For

$$A = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}, \qquad x = (1, 1),$$

the output energy is zero, $\|x\|_2^2 = 2$, and $R^2 = 4$. This rational anchor checks the Schur interface and also illustrates that an upper envelope does not assert response lower bounds.

3 Audit protocol

For each of six windows, the producer constructs the matrix, classifies the four masks, applies all nine controls, and stores the source norm, response gain, Frobenius gain, Schur gain, hybrid gap, occupancy, and active branch. This gives $6 \times 9 \times 4 = 216$ records, of which 198 have nonzero source norm. A reverse-shell independent checker rebuilds the source and matrix under parent hash locks. A mutation stress test changes structural counts, bound flags, branch counts, and firewall fields; every mutation must be rejected.

4 Finite results

Table 1: Hybrid-envelope readout on the declared six-window panel.

mask	nonempty	hybrid occupancy	Frobenius improvement
twin prime	54	0.028830–0.186855	1–1
non-twin prime shift	54	0.010649–0.055850	1–1
prime-power shift	36	1.000000–1.000000	1–1
zero support	54	0.035013–0.041164	1.250245–4.698443

All 216 hybrid inequalities pass. The active-branch census is

$$\#\{\text{Schur records}\} = 54, \qquad \#\{\text{Frobenius records}\} = 162.$$

The global nonempty occupancy range is 0.0106490382–1.0000000000. On the broad twin, non-twin, and zero masks the maximum is 0.1868550366. The Schur branch is consequently useful where the Frobenius column sum is inflated, especially for the zero-support records, but it does not recover alignment information for broad masks.

The strongest positive result is a clean finite, sign-free envelope with a measurable branch improvement. The strongest obstruction is equally clear: support information plus a global row-sum bound still leaves a large response gap on broad masks. The occupancy values are finite observations, not a uniform-in-x or growing-in-N theorem.

5 Route evaluation and scope

The Schur inequality and the hybrid construction are `PROVED_EXACT_FINITE_DECLARED_MODEL`. The replay and zero-violation census are `NUMERICALLY_CERTIFIED_FINITE`; the occupancy and branch ranges are finite numerical observations. Broad-mask factor-five tightness is `REFUTED_SCOPED`. The Session-named official Route-A and Route-B evaluator files are absent, so the local Bridge-B wrapper is fail-closed and does not assert an official route pass.

In particular,

`ARITHMETIC_ADVANCE=NO`, `FIXED_POWER_CREDIT=0`, `FULL_GATE_B=OPEN`.

There is no twin-prime theorem in this release. The reusable structure is the branch-audited combination of a support-local norm and a global induced norm; the unresolved gate is a uniform arithmetic operator estimate.

6 Next question

Since the hybrid cannot explain broad-mask alignment, the next minimal test is structural rather than another generic norm. On a fresh source holdout we will project the twin response away from the span of the non-twin, zero, and prime-power nuisance means, then repeat the test under adversarial controls. This can either reveal a stable residual direction or provide a sharper obstruction to treating raw twin response as isolated arithmetic signal.